2024-10-28-ApplsLinProg

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Slide 1

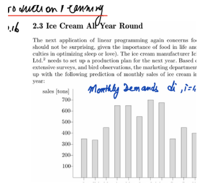
Content

Production Planning
CM&OP.16
2.3 Ice Cream All Year Round
The next application of linear programming again concerns food (which should not be surprising, given the importance of food in life and the difficulties in optimizing sleep or love). The ice cream manufacturer Icicle World Ltd. needs to set up a production plan for the next year. Based on historical data, extensive surveys, and bird observations, the marketing department has come up with the following prediction of monthly sales of ice cream in the next year:
sales [tons] **Monthly demands** $\mathrm{d}_i, i = 1, \ldots, 12$

Description

This slide introduces a linear programming problem concerning the production planning of ice cream for a year. The title "Production Planning" is prominently displayed, along with a reference to a textbook or document ("CM&OP.16"). The main text describes the problem faced by Icicle World Ltd., an ice cream manufacturer, which needs to create a production plan for the next year. The company has gathered data on monthly ice cream sales through historical data, surveys, and even bird observations (a humorous touch). The slide presents a bar chart visualizing the predicted monthly demands (d_i) for ice cream, where i ranges from 1 to 12, representing the months of the year. The chart shows fluctuating demand throughout the year, with higher demand during the warmer months. The context of the course (Linear Programming) is crucial because the problem is framed as a linear programming application, implying that the production plan will be optimized using linear programming techniques. The slide sets the stage for the subsequent slides, which will likely detail the formulation and solution of the linear programming model.

Figures



(a) A bar chart showing the monthly demands for ice cream (in tons) from January to December. The y-axis represents sales in tons, and the x-axis represents the months of the year. The bars show varying demand throughout the year, with peaks in the summer months.

Slide 2

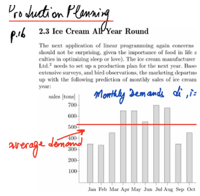
Content

Production Planning
CM&OP.16
2.3 Ice Cream All Year Round
The next application of linear programming again concerns food (which should not be surprising, given the importance of food in life and the difficulties in optimizing sleep or love). The ice cream manufacturer Icicle World Ltd. needs to set up a production plan for the next year. Based on historical data, extensive surveys, and bird observations, the marketing department has come up with the following prediction of monthly sales of ice cream in the next year:
sales [tons] **Monthly demands** $\mathrm{d}_i, i = 1, \ldots, 12$

Description

This slide introduces a linear programming problem focused on production planning for an ice cream manufacturer, Icicle World Ltd. The company needs to determine a yearly production plan based on predicted monthly sales. The prediction is based on historical data, surveys, and even bird observations. A bar chart displays the predicted monthly demands (d_i) for ice cream, where i represents the month (1 to 12). The chart visually shows the fluctuating demand throughout the year, with higher demand during summer months. A horizontal red line indicates the average monthly demand, which is approximately 500 tons. The text "average demand" is handwritten in red next to this line. The context of the course (Linear Programming) is crucial because the problem is framed as a linear programming application, implying that the production plan will be optimized using linear programming techniques. The slide sets the stage for subsequent slides, which will likely detail the formulation and solution of the linear programming model. The reference "CM&OP.16" suggests a textbook or course material.

Figures



(a) A bar chart showing the monthly demands for ice cream (in tons) from January to December. The y-axis represents sales in tons, and the x-axis represents the months of the year. The bars show varying demand throughout the year, with peaks in the summer months. A red horizontal line indicates an average demand, and the word "average demand" is written in red next to the line.

Slide 3

Content

Production Planning
CM&OP.16
(1) X_i - production at month i
(2) S_i - surplus at end of month i
(3) $X_i + S_{i-1} = d_i + S_i$

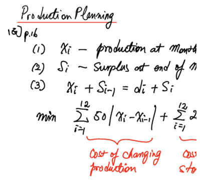
$$\min \sum_{i=1}^{12} 50|X_i - X_{i-1}| + \sum_{i=1}^{12} 20S_i$$

$$\underbrace{\text{Cost of changing production}}_{\text{Cost of storage}}$$

Description

This slide presents a linear programming model for production planning, likely a continuation of the ice cream production problem from the previous slide. The model uses two variables: X_i, representing the production quantity in month i, and S_i, representing the surplus inventory at the end of month i. The equation $X_i + S_{i-1} = d_i + S_i$ represents a mass balance constraint, ensuring that production plus beginning inventory equals demand plus ending inventory for each month. The objective function is to minimize the total cost, which consists of two parts: a cost associated with changes in production from month to month ($\sum_{i=1}^{12} 50|X_i - X_{i-1}|$) and a cost associated with storing surplus inventory ($\sum_{i=1}^{12} 20S_i$). The cost of changing production is 50 per unit change, and the cost of storage is 20 per unit. The summation runs from $i = 1$ to 12, representing the 12 months of the year. The underbraces highlight the two cost components in the objective function. The reference "CM&OP.16" likely refers to a textbook or course material. The context of linear programming is crucial because the slide presents a mathematical model that can be solved using linear programming techniques to find the optimal production plan that minimizes the total cost.

Figures



(a) Handwritten notes describing a linear programming model for production planning. The notes define X_i as the production at month i and S_i as the surplus at the end of month i. A constraint equation $X_i + S_{i-1} = d_i + S_i$ is given, where d_i represents the demand in month i. The objective function is to minimize the sum of the cost of changing production and the cost of storage, represented by $\sum_{i=1}^{12} 50|X_i - X_{i-1}| + \sum_{i=1}^{12} 20S_i$.

Slide 4

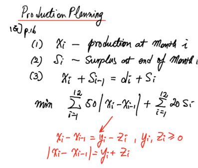
Content

Production Planning
[CM&]P.16
(1) X_i - production at month i
(2) S_i - Surplus at end of month i
(3) $X_i + S_{i-1} = d_i + S_i$
$\min \sum_{i=1}^{12} 50|X_i - X_{i-1}| + \sum_{i=1}^{12} 20S_i$
$X_i - X_{i-1} = y_i - z_i \quad y_i, z_i \geq 0$
$|X_i - X_{i-1}| = y_i + z_i$

Description

This slide details a linear programming model for production planning, likely a continuation from the previous slide's ice cream production problem. It defines X_i as the production quantity in month i and S_i as the surplus inventory at the end of month i. The equation $X_i + S_{i-1} = d_i + S_i$ represents the mass balance constraint for each month. The objective function aims to minimize the total cost, comprising the cost of changing production ($\sum_{i=1}^{12} 50|X_i - X_{i-1}|$) and the cost of storing surplus inventory ($\sum_{i=1}^{12} 20S_i$). Crucially, the slide shows how to linearize the absolute value term $|X_i - X_{i-1}|$ by introducing non-negative auxiliary variables y_i and z_i. The equations $X_i - X_{i-1} = y_i - z_i$ and $|X_i - X_{i-1}| = y_i + z_i$ are added as constraints, transforming the problem into a standard linear program solvable by simplex or interior-point methods. The reference "[CM&]P.16" likely points to a textbook or course material. The context of linear programming is essential because the slide presents a mathematical model that can be solved using linear programming techniques to find the optimal production plan minimizing the total cost. The handwritten nature suggests a lecture or working notes.

Figures



(a) Handwritten notes describing a linear programming model for production planning. The notes define X_i as the production at month i and S_i as the surplus at the end of month i. A constraint equation $X_i + S_{i-1} = d_i + S_i$ is given, where d_i represents the demand in month i. The objective function is to minimize the sum of the cost of changing production and the cost of storage, represented by $\sum_{i=1}^{12} 50|X_i - X_{i-1}| + \sum_{i=1}^{12} 20S_i$. The absolute value in the objective function is linearized using two new variables y_i and z_i, with the constraints $X_i - X_{i-1} = y_i - z_i$ and $|X_i - X_{i-1}| = y_i + z_i$, where $y_i, z_i \geq 0$.

Slide 5

Content

Production Planning
[CM&]P.16

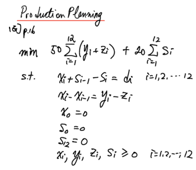
$$\min \;\; 50\sum_{i=1}^{12}(y_i+z_i)+20\sum_{i=1}^{12}S_i$$

s.t. $X_i + S_{i-1} - S_i = d_i \quad i = 1, 2, \ldots, 12$
$X_i - X_{i-1} = y_i - z_i$
$X_0 = 0$
$S_0 = 0$
$S_{12} = 0$
$x_i, y_i, z_i, S_i \geq 0 \quad i = 1, 2, \ldots, 12$

Description

This slide presents the complete linear programming formulation for the ice cream production planning problem. Building upon the previous slides, it shows the final linearized model ready for solution. The objective function minimizes the total cost, which is composed of two parts: $50\sum_{i=1}^{12}(y_i+z_i)$ representing the cost of changing production levels between consecutive months (linearized using y_i and z_i from the previous slide), and $20\sum_{i=1}^{12}S_i$ representing the cost of storing surplus ice cream. The constraints include: a mass balance equation ($X_i + S_{i-1} - S_i = d_i$) ensuring that production plus beginning inventory equals demand plus ending inventory for each month; a linearization constraint ($X_i - X_{i-1} = y_i - z_i$) relating the change in production to the auxiliary variables; and boundary conditions ($X_0 = 0$, $S_0 = 0$, $S_{12} = 0$) specifying initial and final inventory levels. Finally, non-negativity constraints ($x_i, y_i, z_i, S_i \geq 0$) are imposed on all variables. The index i runs from 1 to 12, representing the twelve months of the year. The notation "s.t." stands for "subject to," indicating the constraints of the optimization problem. The context of linear programming is crucial because this slide presents a fully formulated linear program that can be solved using standard linear programming techniques (e.g., simplex method) to obtain the optimal production plan.

Figures



(a) Handwritten notes describing a linear programming model for production planning. The notes define X_i as the production at month i, S_i as the surplus at the end of month i, y_i and z_i as auxiliary variables used in the linearization of the absolute value. The objective function is to minimize the sum of the cost of changing production and the cost of storage, represented by $50\sum_{i=1}^{12}(y_i + z_i) + 20\sum_{i=1}^{12} S_i$. The constraints include a mass balance constraint ($X_i + S_{i-1} - S_i = d_i$), a constraint for linearization ($X_i - X_{i-1} = y_i - z_i$), and boundary conditions for the first and last months ($X_0 = 0$, $S_0 = 0$, $S_{12} = 0$). Non-negativity constraints are imposed on all variables. The index i ranges from 1 to 12, representing the 12 months of the year.

Slide 6

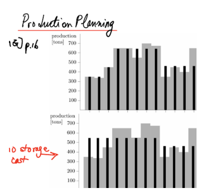
Content

Production Planning
CM&OP.16

Description

This slide presents two bar charts comparing two production planning scenarios for 12 months. The top chart depicts a scenario where storage is allowed, resulting in production levels that don't exactly match demand each month. The bottom chart shows a scenario where there is no storage cost, leading to production levels that closely track demand each month. The y-axis represents production (in tons), ranging from 0 to 700. The x-axis represents the 12 months. Black bars represent the production quantity in each month, while grey bars likely represent the demand for each month. The red arrow and annotation "no storage cost" highlight the key difference between the two scenarios. The context of linear programming is implied because the slide visually compares different production plans, suggesting an optimization problem where the goal is to find the optimal production plan that minimizes costs, potentially including storage costs. The reference "CM&OP.16" likely refers to a textbook or course material.

Figures



(a) Two bar charts showing production levels over 12 months. The top chart shows a scenario with storage, while the bottom chart shows a scenario with no storage cost. The y-axis represents production in tons, ranging from 0 to 700. The x-axis represents the 12 months. Black bars represent production, and grey bars represent demand. A red arrow points from the bottom chart to the right with the text "no storage cost".

Slide 7

Content

Sales Force Planning

[ ]p.209

Jan	390	May	310	Sep	550
Feb	420	Jun	590	Oct	360
Mar	340	Jul	340	Nov	420
Apr	320	Aug	580	Dec	600

TABLE 12.2. Projected labor hours by month.

12.7 Sales Force Planning. A distributor of office equipment finds that the business has seasonal peaks and valleys. The company uses two types of sales persons: (a) regular employees who are employed year-round and cost the company \$17.50/h (fully loaded for benefits and taxes) and (b) temporary employees supplied by an outside agency at a cost of \$25/h. Projections for the number of hours of labor by month for the following year are shown in Table 12.2. Let a_i denote the number of hours of labor needed for month i and let x denote the number of hours per month of labor that will be handled by regular employees. To minimize total labor costs, one needs to solve the following optimization problem:

minimize $\sum_i (25 \max(a_i - x, 0) + 17.50x)$

(a) Show how to reformulate this problem as a linear programming problem.
(b) Solve the problem for the specific data given above.
(c) Use calculus to find a formula giving the optimal value for x.

Description

This slide presents a sales force planning problem. A company uses two types of employees: regular (costing \$17.50/hour) and temporary (costing \$25/hour). The demand for labor hours varies monthly (Table 12.2 provides the data). The goal is to minimize the total labor cost. The initial optimization problem is given as: minimize $\sum_i (25 \max(a_i - x, 0) + 17.50x)$, where a_i is the labor hours needed in month i, and x is the number of hours per month handled by regular employees. The problem is presented as an exercise with three parts: (a) reformulate as a linear program, (b) solve using the given data, and (c) find an optimal x using calculus. The context of linear programming is crucial because part (a) requires transforming the given non-linear problem into a linear programming problem, which is a core concept in the course. The handwritten note "[ ]p.209" likely refers to a page number in a textbook or other course material.

Figures

(a) Handwritten reference [ ]p.209 **(b)** Table 12.2

Slide 8

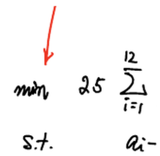
Content

min $25\sum_{i=1}^{12} y_i + (17.5 \times 12)x$
s.t. $a_i - x \leq y_i$
$0 \leq y_i$
$x \geq 0$
$\max(a_i - x, 0) \leq y_i$

Description

This slide shows the reformulated linear programming model for the sales force planning problem from the previous slide. The original non-linear objective function, involving the max function, has been linearized using the auxiliary variable y_i. The objective function minimizes the total labor cost, which is the sum of the cost of temporary employees ($25\sum_{i=1}^{12} y_i$) and the cost of regular employees ($(17.5 \times 12)x$). The constraints ensure that y_i correctly represents the number of temporary employee hours needed in month i. The constraint $a_i - x \leq y_i$ states that the difference between the required hours (a_i) and the hours covered by regular employees (x) must be less than or equal to the hours covered by temporary employees (y_i). The constraints $0 \leq y_i$ and $x \geq 0$ ensure non-negativity. The constraint $\max(a_i - x, 0) \leq y_i$ ensures that y_i is greater than or equal to the maximum of 0 and the difference between required and regular hours, effectively linearizing the original problem. The red arrow highlights the objective function. The context of linear programming is crucial because this slide demonstrates a key technique for transforming a non-linear problem into a linear program, making it solvable using standard linear programming techniques.

Figures



(a) Handwritten notes describing a linear programming model. The notes define x as the number of hours per month of labor that will be handled by regular employees, and y_i as an auxiliary variable. The objective function is to minimize the total labor costs, represented by $25\sum_{i=1}^{12} y_i + (17.5 \times 12)x$. The constraints include $a_i - x \leq y_i$, $0 \leq y_i$, $x \geq 0$, and $\max(a_i - x, 0) \leq y_i$. The index i ranges from 1 to 12, representing the 12 months of the year. A red arrow points to the objective function.